Madhushree Mondal

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About

I'm a B.Tech student in Computer Science and Engineering (IoT, Cyber Security Blockchain Technology) with interests in AI/ML, Computer Vision, Robotics, Blockchain, and IoT. Passionate about building intelligent and secure systems, I have worked on projects involving human tracking, autonomous robotics, and privacy-preserving blockchain applications, with a focus on solving real-world problems through technology.

Education

University of Engineering & Management, Kolkata — *Jul 2024 – Apr 2028*
Bachelor of Technology in CSE (IoT, CS & Blockchain Technology); CGPA: 8.7
SGPA: 1st Sem: 8.74 — 2nd Sem: 8.74 — 3rd Sem: 8.81

Rampurhat Girls High School — *2013–2023*
WBBSE (X): 92% — WBCHSE (XII): 83%

Projects

Blockchain-Based Supply Chain Transparency for Agricultural Produce Website — *2025*
- Developed a blockchain-based agricultural supply chain platform for secure and transparent product tracking.
- Implemented Solidity smart contracts for immutable transaction records and reduced data manipulation risks.
- Built React and Node.js modules to improve stakeholder visibility and traceability.
- **Tech Stack:** *Solidity, Hardhat, React, Node.js, PostgreSQL, Ethers.js* — GitHub

Privacy Layer Using Zero-Knowledge Proofs for Multi-Chain Agricultural Blockchains — *2026*
- Developed a privacy-preserving blockchain application using Zero-Knowledge Proof concepts.
- Enabled confidential compliance verification across multiple blockchain networks.
- Integrated SnarkJS and Circom for improved security, tamper resistance, and data privacy.
- **Tech Stack:** *Solidity, Hardhat, SnarkJS, Circom, React, Node.js, Docker* — GitHub

Human Detection, Direction Tracking & Human-Following Robot — *2026*
- Developed AI-based systems for real-time human detection, directional tracking, and autonomous human following.
- Integrated deep learning and computer vision techniques using OpenCV and Roboflow.
- Implemented Raspberry Pi and IoT sensor modules for intelligent tracking and navigation.
- **Tech Stack:** *Python, OpenCV, Roboflow, Raspberry Pi, NumPy, IoT Sensors* — GitHub

Skills

Languages: Python, C, C++, Java, JavaScript
Core CS: Data Structures and Algorithms, Object-Oriented Programming
AI/ML: Scikit-learn, OpenCV, YOLO, Roboflow
Embedded/IoT: Arduino UNO, Raspberry Pi
Blockchain: Hardhat, Solidity, SnarkJS
Web Technologies: Node.js, HTML, CSS
Databases: MongoDB
Cloud: AWS, Google Cloud
Tools: Jupyter Notebook, Google Colab, AutoCAD, MATLAB, Proteus, ModelSim
Version Control: Git, GitHub

Achievements

- Finalist, **Smart India Hackathon 2025** – Blockchain Technology for Agriculture — Certificate
- NPTEL – *The Joy of Computing using Python* (83%, Elite + Silver) — Certificate

Certifications

- Fundamental Guide to AI for Professionals – Udemy (2025)
- Cyber Security Fundamentals – University of London (2025)
- Information Theory – Coursera (2026)
- Deloitte Certificate in Cyber Security (2026)